

Niramay M. Patel

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Education

Indian Institute of Science Education and Research (IISER), Bhopal *August 2023 – July 2027*
BS in Computer Science CPI: 9.29/10.0

Atmiya Vidya Mandir, Surat, Gujarat *CBSE XII: 94.8%* *Passed in March 2023*

Publications

- **Regulating Internal Evidence Flows for Robust Learning Under Spurious Correlations**
Accepted at ICLR 2026 ([OpenReview](#) [🔗](#))
- **Bias-Aware Dual-Stream Vision Transformer for Fair Skin Lesion Classification**
Submitted to WACV 2027
- **Foundation-Model Guided Pooling for Robust Image Classification Under Background Bias**
Submitted to BMVC 2026
- **Hyperparameter Strategies for Noise-Robust Speech-to-Text**
Submitted to IEEE SLT 2026

Experience

IUSSTF–Viterbi Summer Research Intern (*On site*) *RoboLAND, USC*
Faculty Supervisor: Prof. Feifei Qian; Mentor: Xingjue Liao *May 2026 – July 2026*

- Selected as **one of 15 students nationwide** for the **fully funded Research Internship** at the University of Southern California.
- Built a **salamander-inspired quadruped robot** with a passive flexible spine and an ATI 6-axis force/torque sensor to measure terrain-dependent body dynamics during locomotion.
- Developed MATLAB and MuJoCo simulation pipelines for walk, trot, bound, and swim gaits, including contact-aware servo spine models, virtual sensing, and strain-to-force estimation.
- Designed a **real-time terrain recognition pipeline** from sensor readings using a standardized 42-D gait-cycle descriptor, nearest-centroid inference, and interpretable residual-vibration correction; obtained **70–100% class-wise accuracy** and formulated confidence-gated servo phase adaptation.

Research Intern (*On site*) *VisDom Lab, IISERB*
Professor: Dr. Vinod Kurmi *May 2025 – April 2026*

- Conducted research on **fairness in deep learning**, including debiasing with bias ranking, fairness analysis of the Descript Audio Codec (DAC), and fairness-aware coreset selection.
- Proposed **Evidence-Gated Suppression (EGS)**, a regularizer that tracks class-conditional, confidence-weighted neurons evidence and applies percentile-gated decay to suppress them during training, improving worst-group robustness without requiring group labels.
- Developed a **reweighting framework** that combines per-example hardness and group-level loss signals to improve robustness and fairness in dermatology image classification across Fitzpatrick skin types.

Research Intern (*Remote*) *IIT Roorkee, India*
Prof: Dr. Raksha Sharma *Jan 2025 – June 2026*

- Conducted research to improve the robustness of **speech-to-text models** under real-world noise by designing a controllable procedural noise simulator and analyzing low-dimensional environmental structure via PSD-based spectral representations.
- Performed large-scale robustness experiments using real and simulated noise, studying noise-injected training and evaluation across clean, real-world, and synthetic acoustic conditions.
- Designed a structured **hyperparameter tuning framework** and proposed a principled evaluation protocol to quantify performance stability and robustness across diverse acoustic environments.

Research Intern (*On site*)

Prof: Dr. P. B. Sujit

MOON Lab, IISERB

Dec 2024 – Jan 2025

- Studied core topics in **Computer Vision**, including OpenVINS, MSKCF, SLAM algorithms, and the mathematical foundations of sensor fusion.
- Implemented **Visual Odometry** on the KITTI dataset, building strong theoretical and practical expertise.

Projects

Fairness in DAC

[github repo](#) 

VisDom Lab, June 2025

- Analyzed potential bias in **ParlerTTS** by studying **DAC RVQ codebooks** to examine whether gender-specific patterns appear at the acoustic token level.
- Designed an inspection pipeline to **capture RVQ tokens before decoding** and visualized gender-swapped prompts using **t-SNE**, finding largely uniform token distributions that suggest bias originates upstream rather than in DAC.

Spuriousity Ranking Stability

VisDom Lab, May 2025

- Conducted a robustness study for **Spuriousity Ranking Stability** by analyzing how SEBRA's self-guided bias ranking behaves under noisy labels, random initialization, and dataloader stochasticity, revealing key reproducibility limitations.
- Systematically evaluated ranking consistency and accuracy trade-offs, showing that **data sampling order dominates ranking variance** while the core SEBRA mechanism remains theoretically sound but practically fragile.

Visual Odometry

[github repo](#) 

MOON LAB, December 2024

- Built strong theoretical grounding through extensive reading on **quaternion geometry**, Kalman Filter, error-state modeling, and drone kinematics.
- Estimated the camera's motion from a sequence of images, a crucial task in autonomous driving and SLAM. Successfully reconstructed the **vehicle's trajectory** and evaluated the accuracy using ground truth data.

Certifications

Machine Learning Specialization by Stanford University and DeepLearning.AI

[certificate](#) 

- **Topics covered:** Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommenders, and Reinforcement Learning.

Co-Curricular and Extra-Curricular

- Proposed an **AI based Dynamic Bus route rationalization** model that would be required based on real-time traffic and road parameters for the **Delhi Government** in Smart India Hackathon.
- Secured a rank **under 500** as a College Ambassador for **Techfest, IIT Bombay**.
- Led the **core committee** in organizing a two-day Navaratri event attended by over 1,000 participants, managing end-to-end planning and execution.
- I enjoy playing table tennis, badminton, and cricket to relax during my free time.

Research Interests

My research interests lie at the intersection of **robotics, control systems, and artificial intelligence**. I am particularly interested in developing **dynamic and adaptive robotic systems** that combine model-based control with **neural network architectures and reinforcement learning**. My goal is to improve the robustness, stability, and decision-making capabilities of robots operating in changing environments.